

13.172: LR 2

Evaluateⁿ

Is my model good enough?

Calculate error

$$e_i = \hat{y}_i - y_i;$$

Total error

$$\sum_i \hat{y}_i - y_i$$

This is not good method. As +ve & -ve errors cancel each other.

So we can take either absolute or square

Absolute function is not differentiable & we want differentiable function to calculate Gradient Descent.

Square function is differentiable, so select square.

MSE (Mean square error)

$$= \frac{\sum (\hat{y}_i - y_i)^2}{N}$$

MAE (Mean absolute error)

$$= \frac{\sum |\hat{y}_i - y_i|}{N}$$

Baseline model

Model-I

sq. area	Price
200	40
400	45
300	60
700	80

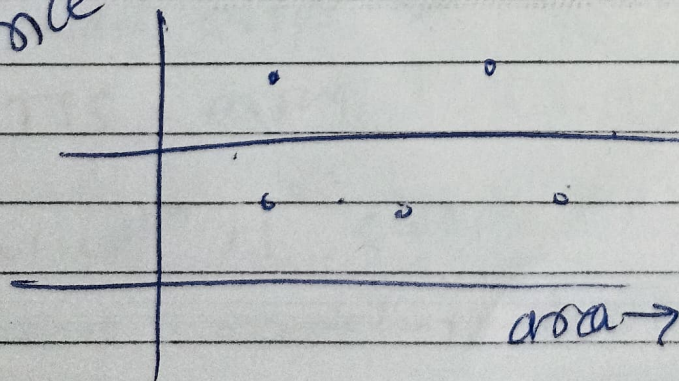
~~avg. price = $\frac{40 + 45 + 60 + 80}{200 + 400 + 300 + 700}$~~

~~per sq.~~

~~so by~~

avg. price = $\frac{40 + 45 + 60 + 80}{4}$

price



→ ~~baseline~~ model I
→ avg. price

Testing

A	P	error ²
40	50	40 10 ²
45	50	5 ²
60	50	10 ²
80	50	30 ²

$$\sum \text{error}^2 = 1125 \rightarrow \text{Total sum of squares}$$
$$\underline{\underline{TSS}}$$
$$= \sum (\text{avg} - \text{actual})^2$$

This is baseline model

This is bad model

model II :- take random m & c

A	P	error ²
40	35	5 ²
45	42	3 ²
60	59	1 ²
80	82	2 ²

~~39~~ (total sum of squared error)
= 39

This is called RSS
Residual sum of error

$$\text{Evaluation} = 1 - \frac{\text{RSS}_{\text{model}}}{\text{TSS}_{\text{avg (baseline)}}}$$

model error ↓
TSS error ↑

↖ ↗
 R^2

evaluation is closer to 1,
means model is better than
baseline.

model error is equal to
baseline error

$$R^2 = 1 - 1 = 0 \quad \text{worst}$$

model error is ~~more than~~ zero
~~baseline error~~

$$R^2 = 1 - 0 = 1 \quad \text{best}$$

model error is more than
baseline error

$$R^2 = -ve$$

Not practically possible

so range of R^2 is $0 \rightarrow 1$

$$\text{Loss} = \text{MSE} = \frac{\sum (A - P)^2}{N}$$

$$= \frac{\sum (y - (\omega^T x + \omega_0))^2}{N}$$

~~$$\frac{d \text{Loss}}{d \omega} = \frac{2}{N} \sum (y - (\omega^T x + \omega_0)) x$$~~

$$= \frac{\sum (y - (\omega_1 x_1 + \omega_2 x_2 + \omega_0))^2}{N}$$

$$\frac{d \text{Loss}}{d \omega_1} = \frac{2}{N} \sum (y - (\omega_1 x_1 + \omega_2 x_2 + \omega_0)) x_1$$

$$= \frac{1}{N} \sum 2 (y - (\omega_1 x_1 + \omega_2 x_2 + \omega_0)) (-x_1)$$

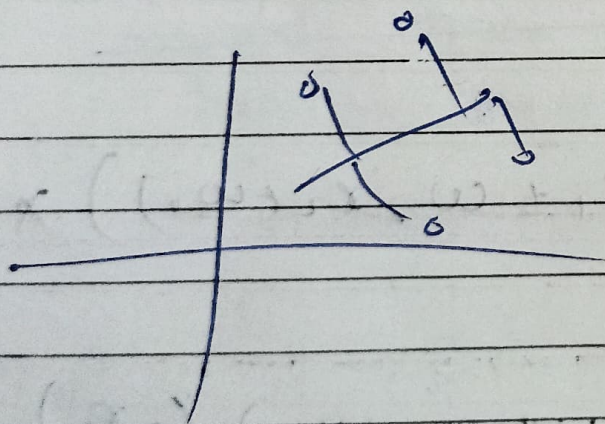
$$\frac{d \text{Loss}}{d \omega_0} = \frac{1}{N} \sum 2 (y - (\omega_1 x_1 + \omega_2 x_2 + \omega_0)) (-1)$$

R^2 is for testing the model after training

GD for training
 $\rightarrow$ MSE

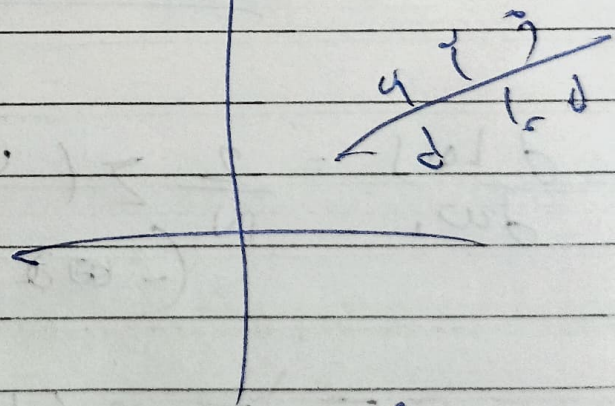
Q. Classification / Latent space & linear regression

Classification



here Σ distance
needs to be max.

linear regression



here Σ distance
needs to be
min.